

Calculating the Fibonacci Sequence

The *Fibonacci sequence* is another classic example for introducing recursion. Mathematically, the Fibonacci sequence of integers begins with the numbers 1 and 1 (or sometimes, 0 and 1). The next number in the sequence is the sum of the previous two numbers. This creates the sequence 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, and so on, forever.

If we call the latest two numbers in the sequence *a* and *b*, you can see in [Figure 2-2](#) how the sequence grows.

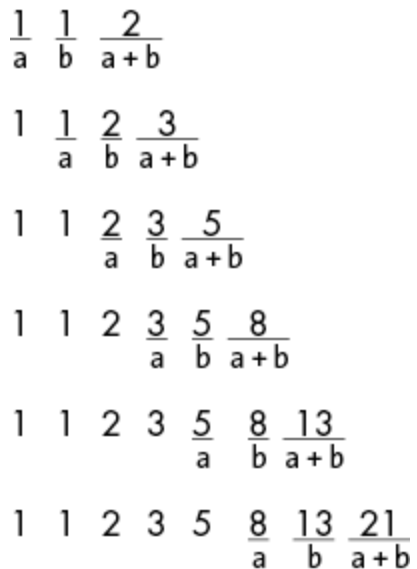


Figure 2-2: Each number of the Fibonacci sequence is the sum of the previous two numbers.

Let's explore some code examples of both the iterative and recursive solutions for generating Fibonacci numbers.

The Iterative Fibonacci Algorithm

The iterative Fibonacci example is straightforward, consisting of a simple `for` loop and two variables, *a* and *b*. This *fibonacciByIteration.py* Python program implements the iterative Fibonacci algorithm:

Python